Distillation + Raw CDF Refinement (v2 Engineered-Feature Architecture)

Mie Spray Postprocessing Project

2026 年 4 月 10 日

Message

Claim: a Stage-2 heteroscedastic teacher trained on fitted curves can be refined toward raw CDF observations using knowledge distillation, with regime-dependent weighting that respects the time-dependent disappearance of usable raw data.

What this deck covers:

- why direct raw-series supervision alone is insufficient
- how CDF coverage regimes partition the time axis
- the distillation loss design: raw NLL + KD KL + onset + anchor + shape priors
- v2-specific changes: A-scaled feature space and physical-space loss
- the student model architecture and training procedure

Motivation: Why Not Just Train on Raw Series?

The problem with raw CDF alone:

- raw CDF observations thin out with time — by $t \sim 3\,$ms many operating conditions have lost $> 70\%$ of their measurements
- fitting only on available raw data produces models that collapse toward zero or become erratic in the late-time regime
- the fitted-curve teacher (Stage-2 NLL) is well-defined everywhere on $[0, 5]\,$ms but was never calibrated against raw observations

Solution: use the teacher's Gaussian predictions as a regularizing prior in time regions where raw data are sparse, and use raw CDF as direct supervision where coverage is strong.

CDF Coverage Regime Labeling

Time is binned into $\Delta t = 0.1$ ms intervals over $[0, 5]$ ms. For each operating-condition group, smoothed raw CDF sample counts determine three regimes:

`raw_reliable` smoothed coverage $\geq 70\%$ of peak — raw CDF is the primary supervision source

`raw_uncertain` coverage has dropped below 70% but above 20% — transitional zone

`teacher_only` coverage below 20% or fewer than 4 raw samples per bin — teacher Gaussian becomes the only supervision

Implementation: `build_time_bin_regimes` produces a per-group, per-bin regime table. Transitions require ≥ 2 consecutive bins to prevent isolated dips from triggering premature regime changes.

Regime-Dependent Supervision Weights

Each grid point receives two independent weight channels:

Regime	Raw weight w_{raw}	KD weight w_{kd}
`raw_reliable`	1.0	0.25
`raw_uncertain`	0.0	1.0
`teacher_only`	0.0	0.75

Design rationale:

- in `raw_reliable`: trust raw data fully, keep mild KD regularization
- in `raw_uncertain`: drop raw supervision entirely, let teacher KL dominate
- in `teacher_only`: slightly reduce KD weight to acknowledge teacher extrapolation uncertainty

Both channels are further modulated by a global time-bin rebalancing weight (inverse-frequency, clipped to $[0.5, 2.0]$).

v2 Adaptation: A-Scaled Feature and Loss Space

Recall from the v2 engineered-feature deck:

$$A = \left(\frac{\Delta P}{\rho_a}\right)^{1/4} d_n^{1/2}, \qquad \hat{\mu}(t) = \mu(t)/A, \qquad \hat{\sigma}(t) = \sigma(t)/A$$

What changes in distillation:

- features are built via `build_feature_matrix_np`, which returns `(features, a_scale, canonical)` — the dataset now carries `a_scale` per sample
- teacher forward pass: $\mu^{\text{phys}}_{\text{teacher}} = A\,\hat{\mu}_{\text{teacher}}$
- student forward pass: $\mu^{\text{phys}}_{\text{student}} = A\,\hat{\mu}_{\text{student}}$
- **all losses are computed in physical space (mm)**: raw NLL, KD KL, anchor penalty, monotonicity and concavity priors

This ensures the loss magnitude is comparable across operating conditions with different A values, and raw CDF targets (which are always in mm) can be used directly.

Student Model Architecture

Output dimension: 3 (vs. teacher's 2)

$$\text{student output} = [\hat{\mu},\ \log \hat{\sigma}^2,\ \text{onset\_logit}]$$

Initialization:

- all shared layers are copied from the teacher checkpoint
- final linear layer: first 2 output rows copied from teacher; onset row initialized to zero (sigmoid output $\to 0.5$)

Feature set: identical to the teacher — `a_only` (5 features) or `a_plus_log_a` (6 features). The student is a pointwise MLP, but batching is sequence-wise so derivative priors operate on contiguous trajectories.

Composite Loss Function

$$\mathcal{L} = \mathcal{L}_{\mathrm{raw}} + \mathcal{L}_{\mathrm{KD}} + \lambda_{\mathrm{onset}}\,\mathcal{L}_{\mathrm{onset}} + \lambda_{\mathrm{anchor}}\,\mathcal{L}_{\mathrm{anchor}} + \lambda_{d_1}\,P_{d_1} + \lambda_{d_2}\,P_{d_2}$$

- $\mathcal{L}_{\mathrm{raw}}$ Gaussian NLL on physical μ, σ^2 vs. raw CDF, weighted by w_{raw}
- $\mathcal{L}_{\mathrm{KD}}$ forward KL divergence $\mathrm{KL}(q_{\mathrm{student}} \| p_{\mathrm{teacher}})$ in physical space, weighted by w_{kd} and a teacher confidence factor $\min(1, \sigma_{\mathrm{ref}}/\sigma_{\mathrm{teacher}})$
- $\mathcal{L}_{\mathrm{onset}}$ BCE on the onset logit vs. a ramp target $\min(t/t_{\mathrm{ramp}}, 1)$, active only for $t \leq 0.2\,\mathrm{ms}$
- $\mathcal{L}_{\mathrm{anchor}}$ quadratic penalty on $\mu(t \approx 0)$, ramping down over the first $0.15\,\mathrm{ms}$
- P_{d_1} monotonicity prior: $\mathrm{ReLU}(-\Delta\mu)^2$
- P_{d_2} concavity prior: $\mathrm{ReLU}(+\Delta^2\mu)^2$, gated to activate after $t \geq 0.5\,\mathrm{ms}$

Teacher Confidence Weighting

The KD loss includes a per-point confidence weight:

$$w_{\text{conf}}(t) = \operatorname{clamp}\left(\frac{\sigma_{\text{ref}}}{\sigma_{\text{teacher}}(t)},\ 0.25,\ 1.0\right), \qquad \sigma_{\text{ref}} = 10\,\text{mm}$$

Purpose: where the teacher's predicted standard deviation is large (i.e. the teacher itself is uncertain), the KD signal is down-weighted. This prevents the student from being pulled toward poorly constrained teacher predictions in extrapolation regions.

The effective KD weight at each grid point is $w_{\text{kd}} \times w_{\text{conf}} \times w_{\text{time\_bin}}$.

Training Procedure

Data split: random $70/15/15$ train/val/test on CDF wide-format samples.

Sampling: `WeightedRandomSampler` with inverse onset-bin frequency, so early-onset and late-onset samples are seen equally often.

Optimization:

- AdamW, $\mathrm{lr} = 3 \times 10^{-3}$, weight decay from teacher config
- gradient clipping at norm 1.0
- early stopping on validation loss, patience $= 20$

Sanity gates before full training:

1. static check: one batch forward pass, verify finite loss and valid teacher confidence bounds
2. tiny overfit check: 30 steps on ≤ 8 samples, confirm loss decreases and onset logit moves toward zero near $t = 0$